

# Data Flow Diagram

## Fit Buddy AI – AI Fitness Plan Generator using Gemini Models

Generative AI Project | FastAPI + Google Gemini

Mariya Darshini (Team Leader), S. Harini, R. Subhashri, S. Mythili

### Main flow

- User → Web Interface (Jinja2 pages) → FastAPI Application
- FastAPI Application → Route / Service logic → Gemini AI Service (or Fallback Generator)
- Gemini AI Service → Generated Plan and Nutrition Tip → FastAPI Application
- FastAPI Application → SQLite Database (users and workout\_plans tables)
- FastAPI Application → Result Page / Users Page

### Data handled

The application processes user-provided details (name, user ID, age, weight, goal, intensity), feedback text, the generated plan and the nutrition tip. The Gemini API key is kept in environment configuration (.env) and is not intended for source-control publication.